

Computer Science B.S. Degree 2025-2026 Curriculum Chart

Math Courses

MATH 19A or MATH 20A

Calculus I
Prerequisites: Math Placement score of 400 or more or Math 3

AM 10

Engr Math Methods I
Prerequisites: Math Placement or Math 3

or

MATH 21

Linear Algebra
Prerequisites: Math Placement or Math 3

MATH 19B or MATH 20B

Calculus II
Prerequisites: Math 19A

AM 30

Multivariate Calculus for Engineers
Prerequisites: AM 10 or Math 21; Math 19B

or

MATH 23A

Vector Calculus
Prerequisites: Math 19B or 20B

CSE 16

Applied Discrete Mathematics
Prerequisites: Math 19A or Math 19B

CSE 40

Machine Learning Basics
Prerequisites: Math 19B or 20B and CSE 30

ECE 30

Engineering Principles of Electronics
Prerequisites: Math 19B

Lower Division Programming Courses

CSE 20

Beginning Programming Python

CSE 12

Computer Systems & Assembly Language
Prerequisites: CSE 20 or CSE 30

CSE 30

Programming Abstractions: Python
Prerequisites: CSE 20; and Math 19A or Math placement score of 400 or more

CSE 13S

Computer Systems and C Programming
Prerequisites: CSE 12

CSE 101

Intro to Data Structures & Algorithms
Prerequisites: CSE 12, CSE 16, CSE 13S, and CSE 30; Math 19B or 20B

Elective Courses

Upper Division
****ELECTIVE**

Upper Division
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Upper Division
****ELECTIVE**

Ψ CSE 195 can satisfy the DC requirement OR an Upper Division Elective, but NOT both.

**Upper Division Electives: 5 credit (or more than 5 credit) upper-division computer science or computer engineering (CSE) courses with a course number between 100-189, or CSE 195, or courses from the Computational Media electives on the back of this chart. Up to two of these electives may be replaced by upper division mathematics electives listed on the back.

CSE 115A, CSE 185S, or CSE 185E cannot be used to satisfy one of the four upper-division elective requirements.

The capstone course can count toward 1 of the 4 required upper division electives.

DC Requirement
(See List Below)

Disciplinary Communication Requirement (DC)

Students of every major must satisfy that major's upper-division Disciplinary Communication (DC) Requirement. The DC Requirement for the Computer Science B.S. is satisfied by completing one of the following courses:

CSE 115A Introduction to Software Engineering

CSE 185E/185S Technical Writing for Computer Science and Engineering

Ψ CSE 195 Senior Thesis

DC courses cannot be used to satisfy any of the 4 Upper Division Electives with the exception of CSE 195.

Upper Division CSE Courses

CSE 120

Computer Architecture
Prerequisites: CSE 12 and CSE 13S

CSE 114A

Foundations of Programming Languages
Prerequisites: CSE 101

CSE 130

Principles of Computer Systems Design
Prerequisites: CSE 12 and CSE 101

CSE 101M

Mathematical Thinking for Computer Science
Prerequisites: CSE 101

CSE 102

Analysis of Algorithms
Prerequisites: CSE 101M

or

CSE 103

Computational Models
Prerequisites: CSE 101M

CSE 107

Probability and Statistics
Prerequisites: CSE 16 and Math 23A or AM 30

or

STAT 131

Intro to Probability Theory
Prerequisites: Math 19B or Math 20B. (AM 30 or Math 23B recommended)

Many upper division Computer Science courses are restricted to enrollment by declared Computer Science majors during first-pass or priority enrollment.

Capstone Courses

Many Capstone course options require additional prerequisites not already required in major requirements. Advance planning is crucial.

CSE 110B Fundamentals of Compiler Design II
CSE 115C Software Design Project II
CSE 115D Software Design Project - Accelerated
CSE 121 Embedded System Design
CSE 134 Embedded Operating Systems
CSE 138 Distributed Systems
CSE 140 Artificial Intelligence
CSE 143 Introduction to Natural Language Processing
CSE 144 Applied Machine Learning
CSE 145 Introduction to Data Mining
CSE 156/L Network Programming / Lab
CSE 157 Internet of Things
CSE 160 Introduction to Computer Graphics / Lab
CSE 161/L Introduction to Data Visualization / Lab
CSE 162/L Advanced Computer Graphics and Animation / Lab
CSE 163 Data Programming for Visualization
CSE 168 Introduction to Augmented Reality and Virtual Reality
CSE 181 Database Systems II
CSE 183 Web Applications
CSE 184 Data Wrangling and Web Scraping
CSE 187 Full Stack Web Development II
CMPM 172 Game Production Studio

These courses can be used to satisfy Upper Division Electives.

Comprehensive Requirement - Students have two options to fulfill the Computer Science exit requirement:

1. Pass one of the Capstone Courses
2. Successfully complete a Senior Thesis.

Disciplinary Communication Requirement - Students have two options to fulfill the DC requirement:

1. Pass one of the Disciplinary Communication Courses
2. Successfully complete a Senior Thesis.

Computer Science B.S. Degree 2025-2026 Curriculum Chart

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Mathematics Electives List

AM 114 Introduction to Dynamical Systems
AM 147 Computational Methods and Applications
AM 148 GPU Programming for Scientific Computations
AM 160 Introduction to Scientific Machine Learning
MATH 110 Introduction to Number Theory
MATH 115 Graph Theory
MATH 116 Combinatorics
MATH 117 Advanced Linear Algebra
MATH 118 Advanced Number Theory
MATH 134 Cryptography
MATH 145/L Introductory Chaos Theory / Lab
MATH 148 Numerical Analysis
MATH 160 Mathematical Logic I
MATH 161 Mathematical Logic II
STAT 132 Classical and Bayesian Inference
One of the following combinations: [PHYS 5A and PHYS 5B] OR [PHYS 5A and PHYS 5C] OR [PHYS 6A and PHYS 6B] OR [PHYS 6A and PHYS 6C]***

Computational Media Electives List

CMPM 120 Game Development Experience
CMPM 131 User Experience for Interactive Media
CMPM 146 Game AI
CMPM 163 Game Graphics and Real Time Rendering
CMPM 164/L Game Engines / Lab
CMPM 171 Game Design Studio
CMPM 172 Game Production Studio

- All courses being applied to requirements for the Computer Science major must be taken for a letter grade. Grades of P will not count toward major requirements.
- Courses in which you receive a grade of C-, D+, D, or D- earn credit toward graduation, but cannot be used to satisfy a major requirement or a general education requirement, and cannot satisfy a prerequisite for another course.
- Shaded boxes represent major qualification courses.
- Many graduate courses can also be used to satisfy electives; however, students will need instructor and department approval.

*** Physics courses have co-requisite labs required for enrollment. These associated labs are not part of the Computer Science B.S. major requirements.